

CHAPTER I

INTRODUCTION

1.1 Rationale of the Study

In higher education, teachers are expected to create an assessment that is fair, valid, and directly related to the teaching of the subject (Fives & DiDonato-Barnes, 2018). The Table of Specifications (TOS) is a common instrument used for this purpose which serves as a kind of blueprint of the exam format for mapping questions to learning objectives and cognitive levels (Fives & DiDonato-Barnes, 2018). A TOS, if well-designed, leads to the correct assessment of understanding and specifically to the appropriate cognitive load (Fives & Barnes, 2018).

In the Philippines, there is a greater need for highly structured assessment because of the flexibility of learning directions as established in CHED CMO No. 4, s. 2020 (CHED, 2020). The move of institutions into flexible learning settings is leading to greater demands for faculty to implement valid, reliable, course-based assessments in multiple modalities.

But the manual setting up of a TOS is a very time-consuming administrative process. Teachers need to determine item difficulty levels, determine how to map cognitive limitations in accordance with Bloom's Taxonomy, and match teaching content to assessment outcomes (Fives & DiDonato-Barnes, 2018). The mechanical flow makes faculty work more complex and time consuming, taking away time from their main teaching responsibilities. A study by EDCOM 2 (2024) indicated that the average working hours of the educators in the Philippines for the week of 2023 are

52 hours, with many exceeding 60 hours, especially for other administrative tasks that are not directly related to teaching.

Although there are different types of educational technologies, there is a lack of automation in the context of exam building guided by the TOS. While Learning Management Systems (LMS) like Moodle are well suited for delivering assessments, they do not come with built-in tools to support teachers in creating exams from a structured TOS blueprint (Gamage et al., 2022). On the other hand, general-purpose Artificial Intelligence (AI) tools can quickly create questions but tend to produce inaccurate content that does not align with the actual syllabus (Augenstein et al., 2024). The fact that these models do not have a strict factual basis of the instructor's concrete resources leads to products that do not necessarily represent the factual topics of the instructional process (Gorgun & Bulut, 2024). In addition, the tools currently available mostly support digital test execution, with little support for the creation of formatted, printable examinations.

In order to overcome these systematical limitations, the study tries to introduce a new approach called Nexam: A RAG and OMR Based System for Syllabus Driven TOS Configuration and Answer Sheet Checking. The platform prepares objective questions based on the teaching material uploaded by the Faculty. It has a human-in-the-loop workflow, meaning that the instructor can review, edit and explicitly approve all created items before they are added to the central question bank. This architecture allows for the educator to be the main driver for the assessment process. The net result: Exam prep tools are designed to minimize workload and maximize pedagogical organization, topic alignment, and printable and LMS-based deployment options.

1.2 Purpose and Project Description

The purpose of this project is to automate the construction of Table of Specifications (TOS)-aligned examinations for higher education faculty, reducing manual workload and ensuring that assessments are valid, structured, and grounded in actual instructional materials through AI-assisted generation and Optical Mark Recognition (OMR)-based checking.

The project will:

- Parse uploaded course syllabi to automatically generate a TOS blueprint with proportional item weights and Bloom's Taxonomy distribution.
- Generate objective-type draft questions grounded exclusively in instructor-uploaded materials using Retrieval-Augmented Generation (RAG) to prevent hallucination and ensure content validity.
- Implement a mandatory human-in-the-loop review workflow requiring instructors to evaluate, edit, and approve every AI-generated question before it enters the question bank.
- Detect and flag near-duplicate questions through semantic similarity checking before saving to the centralized question bank.
- Produce print-ready exam sets with answer keys and TOS summary reports, and export question banks in GIFT format for Moodle and XML format for Canvas.
- Enable mobile-based OMR scanning of completed answer sheets to automatically compute scores and sync results to a web-based performance dashboard.

The system will function as a multi-platform solution accessible via web and mobile applications, integrating syllabus parsing, AI-assisted question generation, question bank management, dual-format exam output, and automated answer sheet checking into a single, centralized platform.

1.3 Objectives of the Study

General Objective

Nexam simplifies the examination process by providing an assisted TOS builder where instructors assign item numbers to each cognitive level, AI-powered question drafting based on Bloom's Taxonomy, and flexible delivery that supports both print-ready formats and Learning Management System (LMS) integration.

Specific Objectives

Specifically, this study aims to:

1. Develop a mobile-responsive web platform and a companion mobile application accessible to faculty.
2. Create a syllabus upload module which extracts topic names, instructional hours, and learning outcomes from a faculty-uploaded syllabus and automatically generates a Table of Specifications (TOS) blueprint with computed item weights and number of questions per learning level based on Bloom's Taxonomy.
3. Create a question drafting module that will produce objective-type questions from materials the instructor uploads, based on the finalized TOS blueprint.

4. Create a question review and approval process that requires instructors to review, edit, and approve all questions generated by the system before they are stored in the question bank.
5. Create a similarity checking feature to identify near-duplicate questions before they are added to the question bank via semantic comparison.
6. Create a school-wide question bank that is stored and readily accessible by subject, topic, cognitive level, and TOS alignment, with the aim of reusing questions throughout the school term.
7. Create printable exam sets matching the TOS, answer key sets, and TOS summary reports, and to support export of the question bank to GIFT format for Moodle and XML format for Canvas.
8. Develop a mobile OMR module that will scan completed answer sheets using the device camera, calculate marks using Optical Mark Recognition, and sync results to the web platform.
9. Develop a performance dashboard that will show individual scores, class averages, and item-level response distributions from scanned OMR results.
10. Evaluate the performance of the AI components using standard classification metrics including Accuracy, Precision, Recall, and F1-Score.

User Evaluation

User Evaluation will follow the ISO/IEC 25010 software quality standards, which provide a structured basis for measuring how well a system meets the needs of its actual users. University instructors will participate in the evaluation through structured surveys and hands-on usage testing.

1.4 Significance of the Study

This study is beneficial for educational institutions, as it proposes a system to generate exam questions from the materials provided by the instructors based on AI with questions categorized according to Bloom's Taxonomy levels. The system generates an exam based on a Table of Specifications (TOS) to determine how test items will be distributed and can produce print-ready test papers or export to an LMS. It assists with essential test preparation components, saving instructors time and effort, maintaining consistency in assessments, and consolidating exam data in a single location.

Instructors. Saves considerable time and effort preparing exams. Rather than calculating item distributions manually and creating questions from scratch, instructors upload their course material to the TOS and set it up by typing in the names of topics and hours of instruction and assigning item numbers to each cognitive level. Then the system generates a pool of questions to be reviewed, edited and approved by instructors. Over time , more questions become validated and added to the central question bank . So the amount of work involved in creating new exams each term keeps getting lower .

Furthermore, this study serves as a baseline guide for developers in integrating AI text extraction, constraint mapping and Optical Mark Recognition (OMR) to develop robust educational testing tools.

1.5 Scope and Limitations

The project focuses on developing Nexam, a web-based and mobile-supported assessment system that will be piloted and deployed at Davao Oriental State University. It will help instructors create examinations using a Table of Specifications (TOS),

AI-assisted question generation grounded in instructor-provided materials, semantic similarity checking, print-ready examination output, LMS-compatible export files, and mobile OMR-based answer-sheet checking.

Scope:

- The system will accept an uploaded course syllabus in PDF or DOCX format and will extract topic names, instructional hours, and learning outcomes to automatically generate the initial TOS blueprint. Item weights will be computed proportionally based on instructional hours, and question counts will be distributed across the six levels of Bloom's Taxonomy.
- In addition to the syllabus, instructors will be able to upload supplementary instructional materials including PDF documents, DOCX files, presentation files, website URLs, and YouTube video transcripts. The system will extract text from these materials and use them as the source content for question drafting.
- The system will draft objective-type questions only specifically multiple-choice, true/false, matching type, and fill-in-the-blank grounded in the instructor's uploaded instructional materials and aligned to the topic and cognitive level specified in the finalized TOS.
- Every system-drafted question will go through a mandatory instructor review process. Instructors will evaluate, edit, and approve each question before it is saved. No question will enter the question bank without the instructor's explicit approval.
- The system will screen every question for duplicates or near-duplicates using semantic vector comparison against all existing approved questions before

saving. Questions flagged as potential duplicates will be presented to the instructor for a final decision before proceeding.

- The question bank will store all approved questions organized by subject, topic, cognitive level, and TOS alignment. Instructors will be able to retrieve and reuse approved questions across different school terms without recreating them from scratch.
- The system will produce print-ready exam sets Set A and Set B with matching structure, difficulty distribution, and TOS alignment. Each set will include a formatted answer key and a TOS summary report. These outputs will be available as downloadable PDF files.
- The system will export the question bank in GIFT format compatible with Moodle and in XML format compatible with Canvas, following the standard import specifications of each platform. Export files will be ready for manual upload into the respective LMS without additional reformatting.
- The mobile application will generate a printable OMR answer sheet paired to each exam set. Each answer sheet will contain one bubble field per item and an embedded QR code that encodes the exam ID and links the sheet to its corresponding answer key stored in the system.
- The mobile application will allow instructors to scan completed OMR answer sheets using the device camera. The application will decode the QR code to retrieve the correct answer key, detect filled responses through Optical Mark Recognition, compute item-level and total scores, and transmit results to the web platform.

- The web platform will aggregate OMR scan results and display them on the instructor's performance dashboard, showing individual student scores, class averages, and item-level response distributions per exam.

Limitations:

- The system will not include a student-facing portal, live online exam delivery, or browser-based exam monitoring of any kind.
- The system will not generate or assist in grading subjective question formats, including essays, diagram labeling, coding exercises, and mathematical problem sets.
- Syllabus parsing requires a clearly formatted digital document. The system will not be able to accurately extract content from handwritten syllabi, scanned image-based PDFs without selectable text, or poorly structured documents.
- OMR scanning accuracy will depend on print quality, ambient lighting, and the completeness of student responses. Partially filled, smudged, or ambiguous marks may not be detected correctly and will require manual correction by the instructor through the application's review interface.
- OMR checking will be limited to the objective question types supported by the system. Open-ended, essay-type, or diagram-based responses cannot be scanned or graded through the OMR module.
- YouTube video processing will require videos with available captions or auto-generated transcripts. Videos without captions will not be usable as source material for question drafting.

- Website URL processing will work best with standard, text-based educational pages. Highly dynamic or JavaScript-heavy pages may not be fully extracted by the system's HTML parsing tools.
- The system will require a stable internet connection for question generation, LMS file export, and OMR score syncing. Core features will not function in offline conditions.
- The system will not connect directly to external LMS platforms. Transferring question banks to Moodle or Canvas will require the instructor to manually upload the exported GIFT or XML file into the respective platform.
- The performance dashboard will be limited to basic output summaries including scores, averages, and correct response rates per item. Predictive analytics, learning progression tracking, and longitudinal performance reporting are outside the scope of this study.

1.6 Conceptual Framework

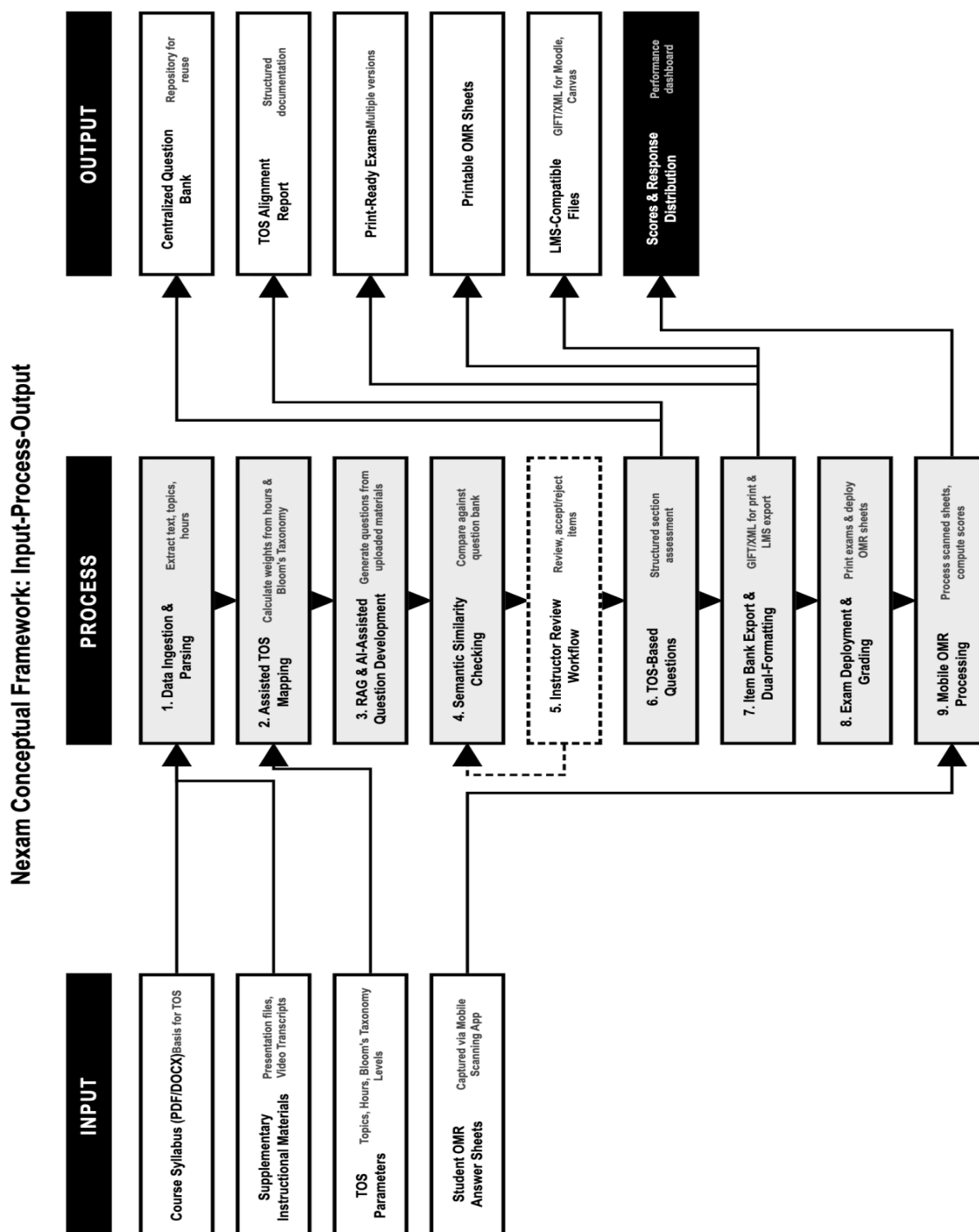


Figure 1 Conceptual Diagram of the Study

The Conceptual Framework

The conceptual framework presents how Nexam transforms instructor-provided academic materials into reviewed, reusable, and measurable assessment outputs. The input consists of the course syllabus, instructional hours, learning outcomes, supplementary course materials, finalized TOS parameters, existing approved questions, and completed OMR answer sheets. These inputs establish the instructional basis for question generation and the reference collection used for duplicate detection.

The process begins when the system extracts and organizes content from the uploaded syllabus and instructional materials. It computes topic weights and the required number of items for each cognitive level, after which Retrieval-Augmented Generation drafts objective-type questions using only the retrieved course content. Every drafted question is then converted into a semantic vector and compared with the vectors of approved questions already stored in the question bank. The cosine-similarity score is evaluated against the study's validated threshold: a score at or above the threshold is categorized as Similar and is flagged for instructor review, while a score below the threshold is categorized as Far Similar and may proceed to the normal review workflow. The instructor remains the final decision-maker and may edit, approve, reject, or retain a flagged item.

After instructor approval, accepted questions are stored in the centralized question bank and assembled according to the finalized TOS. The system then generates the required examination materials and supports mobile OMR processing of completed answer sheets. This sequence makes the similarity-checking mechanism

explicit: question text is embedded, compared against approved items, classified according to the threshold, and routed to the instructor before storage.

The outputs of the framework are an instructor-approved question bank, a similarity result for each drafted item, TOS alignment reports, print-ready examination sets with answer keys, printable OMR sheets, LMS-compatible export files, and a performance dashboard containing scores and item-level summaries. These outputs demonstrate that Nexam does not merely generate questions; it produces reviewed assessment resources with documented alignment, duplicate-control results, and usable examination and scoring artifacts.

1.7 Definition of Terms

For a better understanding of the terms used in the study, the following are defined:

AI Hallucination	When AI invents incorrect information. The system prevents this by restricting generation exclusively to instructor-uploaded materials.
Bloom's Taxonomy	A framework used to classify the cognitive difficulty of test questions.
Cognitive Level	The specific level of thinking required for a student to answer a question.
Content Validity	A measure of whether an exam accurately

tests what students were actually taught in class.

Data Ingestion & Extraction

The process of pulling and cleaning text from uploaded files so the AI can use it to create questions.

Dual-Format Delivery

The system's ability to output exams as both print-ready papers and LMS-compatible digital files.

Flexible Learning

The system's ability to generate both printable offline exam papers (with corresponding OMR sheets) and LMS-compatible export files (e.g., GIFT/XML).

Learning Management System (LMS)

The system's ability to output exams as both print-ready papers and LMS-compatible digital files.

Mobile OMR (Optical Mark Recognition)

A mobile application feature that scans printed answer sheets to automatically compute scores

Multimodal Data Extraction

The system's ability to pull text from varied sources, including PDFs, DOCX, websites,

	and video transcripts.
Objective Formats	The system's ability to pull text from varied sources, including PDFs, DOCX, websites, and video transcripts.
Question Bank	A part of the system where test questions are stored and organized for use in exams. .
Retrieval-Augmented Generation (RAG)	An AI technique that guides question generation using instructor-provided text as the primary source.
Semantic Similarity Checking	An algorithm that flags duplicate questions by comparing their underlying meaning rather than just exact wording.
Table of Specifications (TOS)	A test blueprint dictating item distribution per topic and cognitive level, which the system automatically computes from the uploaded syllabus.

CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Literature Map

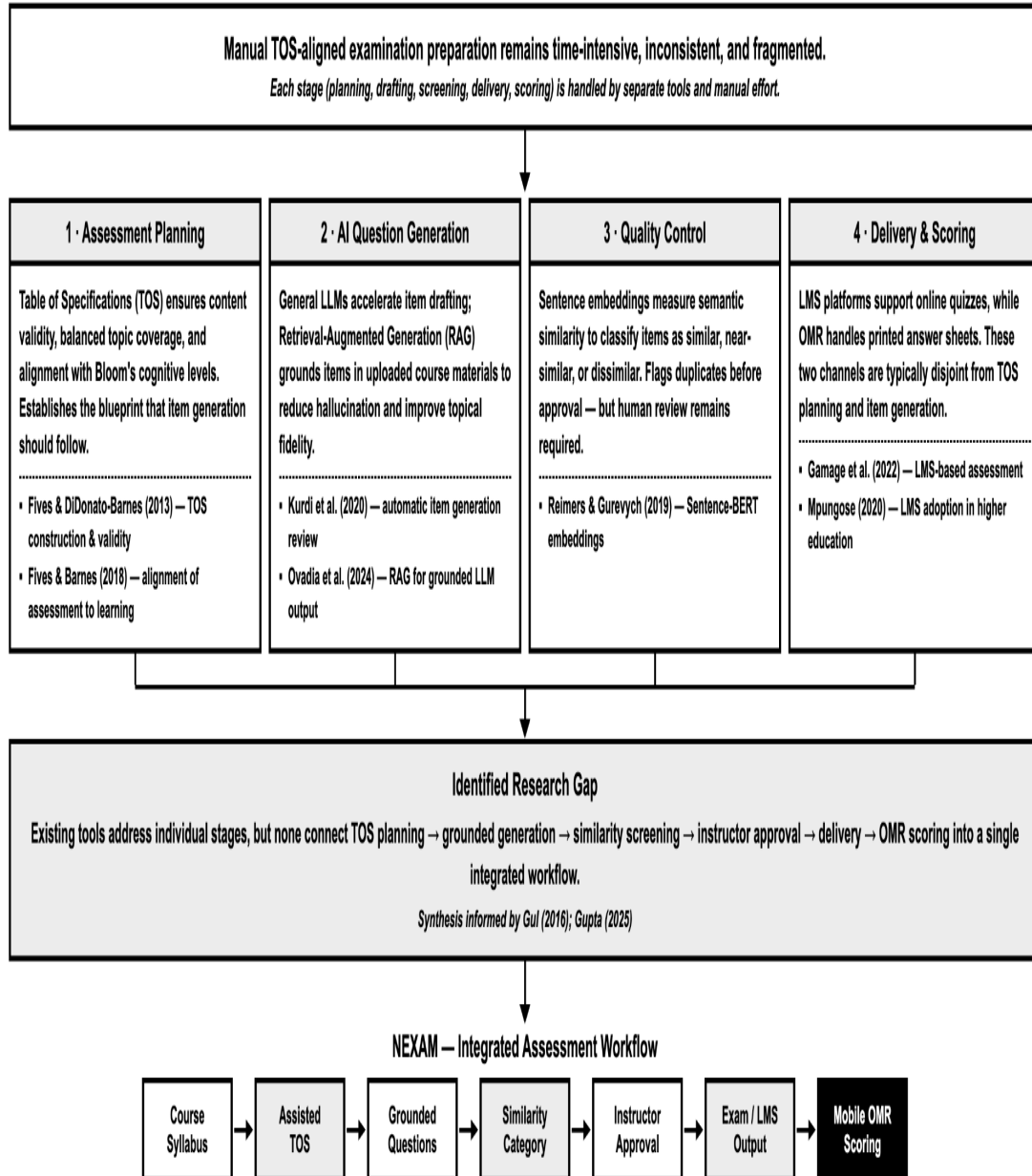


Figure 2 Literature Map of the Nexam Study

2.2 Related Literature

Table of Specifications and Assessment Quality

A Table of Specifications (TOS), commonly referred to as a test blueprint, is a widely used tool in educational assessment. It works as a two-dimensional chart that connects specific course content with learning objectives and expected student outcomes, giving teachers a clear and organized basis for building their exams (Fives & DiDonato-Barnes, 2013). Rather than relying on guesswork or personal preference when selecting test items, a TOS guides the construction process in a systematic and deliberate way, helping ensure that exam questions are distributed across topics and cognitive levels (Fives & DiDonato-Barnes, 2013; Gul, 2016). This structure reduces the chances of some content areas being overrepresented while others are underrepresented, making assessments more transparent for students and providing teachers with a clear basis for showing that their exams reflect what was taught (Fives & DiDonato-Barnes, 2013; Fives & Barnes, 2018).

The value of a TOS in producing valid exams comes from its ability to support test content validity and response process validity (Fives & Barnes, 2018). Test content validity is supported when an exam represents the range of topics covered during instruction, with the amount of class time spent on each topic influencing how much weight it carries in the exam (Fives & DiDonato-Barnes, 2013; Gul, 2016). Response process validity relates to whether the cognitive demands required to answer exam questions match the type of thinking that was practiced during instruction, which is why TOS construction often uses frameworks like Bloom's Taxonomy to ensure this alignment (Fives & Barnes, 2018; Gul, 2016). Despite its importance, building a TOS

manually tends to be time-consuming and prone to error. Instructors must carefully calculate how many questions each topic should receive based on instructional hours and then distribute those questions across multiple levels of Bloom's Taxonomy, all by hand. This gap between the recognized value of a TOS and the effort required to construct one manually is one of the core problems that Nexam is designed to help address, specifically by calculating item distributions automatically once the instructor has entered their topic names and instructional hours, so that the computational work is handled by the system rather than the instructor.

Bloom's Taxonomy in Assessment Design

The alignment of exam questions with the level of thinking expected from students is an important part of good assessment design (Fives & Barnes, 2018). Educators often use frameworks such as the revised Bloom's Taxonomy to group learning objectives into levels, from simple recall to more complex tasks like analysis and creation (Banujan et al., 2023). Each test question is expected to match a specific level of thinking so that the exam can better reflect what students are supposed to learn (Fives & DiDonato-Barnes, 2013). In practice, a Table of Specifications (TOS) is used to organize topics and match them with the appropriate cognitive levels (Fives & Barnes, 2018).

However, applying this process manually can be inconsistent and prone to error (Banujan et al., 2023). Classifying questions by hand takes time and can be subjective, as instructors may find it difficult to clearly distinguish between the different levels of Bloom's Taxonomy. This can lead to mistakes in classification and a tendency to focus more on lower-level questions that test simple recall (Banujan et al., 2023). To address

these challenges, some studies have explored the use of machine learning models to assist in classifying questions based on their cognitive level, offering a more consistent approach compared to manual classification (Banujan et al., 2023).

This gap between the importance of cognitive alignment and the difficulty of applying it consistently by hand is one of the areas that Nexam is designed to help address. In the system, instructors define how questions are distributed across Bloom's Taxonomy levels when they set up their Table of Specifications, specifying what percentage of items per topic should fall under each cognitive level. The AI generation module then produces draft questions intended to match these specified levels, and each generated item is presented to the instructor with a suggested Bloom's Taxonomy level tag during the review process. The instructor reviews, confirms, or adjusts the tag before approving the item for the question bank, meaning the final cognitive classification of every item reflects the instructor's judgment rather than being applied automatically without oversight.

Administrative Burden on Educators

Teaching has always involved more than just standing in front of a classroom, but in recent years the gap between what teachers are contracted to do and what they actually end up doing has grown into a serious problem. Studies show that teachers commonly work between 35 and 50 hours per week, often going beyond their required schedules just to keep up with responsibilities that have nothing to do with teaching (TNTP, 2024). A national survey found that educators spend around 53 hours a week on work-related tasks on average, yet less than half of that time is actually spent directly teaching students (Steiner et al., 2023). In the Philippines, the situation is just as difficult,

with public school teachers averaging 52 working hours per week and nearly 27 percent reporting more than 60 hours of work per week (EDCOM 2, 2024). Much of this extra time goes toward non-teaching tasks like lesson preparation, grading, and administrative paperwork, and the toll it takes is significant. By 2024, over 60 percent of teachers reported experiencing burnout, with unmanageable workloads identified as the main reason (TNTP, 2024).

The weight of these responsibilities does not just cause physical exhaustion. It also affects how teachers feel about their jobs. Research shows a clear connection between the amount of time spent on administrative work and lower levels of job satisfaction, particularly when it comes to the overall work environment (Toropova et al., 2021). When teachers are constantly pulled away from teaching to deal with paperwork and compliance requirements, it creates a sense of conflict between what they are actually there to do and what they are being asked to do instead (Toropova et al., 2021). This problem is especially common in early childhood education, where managing student records and submitting learning outcome reports are among the leading causes of emotional exhaustion and loss of motivation (Fuentes-Vilugrón et al., 2025). When administrative work regularly spills into personal time, many teachers eventually decide to leave the profession altogether (Fuentes-Vilugrón et al., 2025). On top of that, the pressure of rushing through reports and non-standardized documentation takes time away from actual teaching, which ultimately affects the quality of education that students receive (EDCOM 2, 2024).

These realities point to a clear need for automation in education. Research suggests that teachers who use AI tools regularly save an average of nearly six hours per week, giving them more time to focus on their students rather than their paperwork

(Gallup & Walton Family Foundation, 2025). AI tools are increasingly being recognized as practical solutions for reducing teacher workloads by handling repetitive tasks like grading, attendance tracking, and resource preparation (Gallup & Walton Family Foundation, 2025). Automated systems have also been shown to produce real efficiency gains while generating data that helps teachers better understand and support their students (Gallup & Walton Family Foundation, 2025). This growing body of evidence supports the development of Nexam, which directly targets one of the most time-consuming tasks in higher education the manual construction of TOS-aligned examinations and addresses it by handling the most repetitive and error-prone parts of the process on the instructor's behalf. Rather than replacing the instructor's role, the system is designed to reduce the time spent on mechanical tasks such as calculating item distributions and drafting questions from scratch, so that instructors can focus their available time on reviewing content and supporting their students. The time savings analysis conducted for this study estimates that Nexam could recover up to 1,230 instructor hours annually across 15 instructors through exam construction assistance alone, a figure grounded specifically in the workflows the system is designed to support.

Learning Management Systems and the Absence of TOS-Aligned Automation

Learning Management Systems (LMS) are now widely used in universities for delivering courses and managing assessments. Moodle is one of the most commonly used platforms, growing from 78 million users in 2015 to over 294 million by 2021, showing how widely it has been adopted by educational institutions (Gamage et al., 2022). It provides tools such as course management, file uploads, grade tracking, and basic quiz creation, making it suitable for everyday teaching needs (Gamage et al., 2022). However, studies indicate that many institutions mainly use these basic features,

such as uploading materials and sharing content, while other capabilities of the platform are less frequently utilized (Mpungose, 2020).

A key limitation, in relation to this study, is that Moodle does not natively support exam construction based on a Table of Specifications (TOS). Instructors can create question banks and build quizzes, but these processes are done manually. Question banks act as organized collections of test items, usually grouped by topic or difficulty level, which allow instructors to store and reuse questions. However, the system does not automatically determine how many questions each topic should have based on instructional time, distribute items across Bloom's Taxonomy levels, or generate questions directly from instructor-provided materials. Gamage et al. (2022) further note in their systematic review that while Moodle supports assessment organization and quiz creation, building these functions still relies on manual input from instructors, and automated support for structured exam planning remains limited.

While some third-party plugins introduce AI-related features into Moodle, these additions do not constitute an integrated workflow, as the platform itself does not natively support TOS-guided planning or structured exam preparation (Gamage et al., 2022). These observations suggest a gap in current LMS platforms. In response, Nexam is designed to combine TOS-based planning, AI-assisted question generation, and flexible exam output within a single system, offering a more structured approach to exam creation.

Automated Question Generation and Its Limitations

Automated Question Generation (AQG) refers to the use of computer-based tools to create exam questions without requiring manual input from the instructor at every step

(Gorgun & Bulut, 2024). Research confirms that manually writing high-quality exam questions is a complex process that requires professional training, substantial experience, and significant time, all of which are resources that most instructors cannot afford to dedicate entirely to test preparation (Kurdi et al., 2020). For teachers who are already managing heavy administrative workloads, AQG has become an appealing solution to reduce the time and effort that exam construction demands (Gorgun & Bulut, 2024). General-purpose AI tools like ChatGPT and Google Gemini have become the most commonly used options for this purpose, given how accessible and easy to use they are. However, these tools were never specifically designed to handle the structured workflow that professional exam construction actually requires, and their limitations become clear when they are applied in a real educational assessment context (Akdeniz et al., 2025).

The more significant issue with using general AI tools for exam generation is not that they cannot read uploaded materials, because both ChatGPT and Gemini do allow file uploads and can generate questions from them. The real problem is what happens after the questions are generated. There is no built-in system in these tools that automatically calculates how many items each topic should receive based on instructional hours, nor is there any mechanism that distributes those items across the different levels of Bloom's Taxonomy as a proper TOS requires (Gorgun & Bulut, 2024; Akdeniz et al., 2025). Research specifically examining how well AI tools align with Bloom's Taxonomy found that models like ChatGPT and Gemini perform adequately at lower cognitive levels such as remembering and understanding, but show significant variation and weaker performance at higher-order levels like analysis and evaluation, meaning that without structured guidance, AI-generated questions tend to cluster around

surface-level recall tasks rather than reflecting the full cognitive range a well-designed exam requires (Akdeniz et al., 2025; Rokhshad et al., 2025). These tools also have no mechanism to check whether a newly generated question already exists in a teacher's previous exams, meaning repeated items can enter the question bank undetected. . Everything the AI produces stays within the chat session itself, with no organized item bank, no automatic TOS report, and no structured export option that produces a print-ready exam or an LMS-compatible file. These gaps are precisely what Nexam is designed to help address, by bringing TOS-guided item distribution, AI-assisted question drafting from instructor-uploaded materials, similarity checking against existing items, structured item bank storage with mandatory instructor review and approval, and dual-format exam output together into a single purpose-built platform. Crucially, no AI-generated question in Nexam enters the item bank or reaches a student without first passing through the instructor's review and approval, ensuring that the instructor's judgment remains the final checkpoint in the assessment construction process.

Retrieval-Augmented Generation and Its Role in Reliable Exam Construction

Retrieval-Augmented Generation (RAG) has emerged as one of the most reliable technical approaches for reducing the problem of AI hallucination by grounding the output of large language models in specific, verified source materials rather than broad internet data (Gupta, 2025). Unlike standard AI tools that generate responses based purely on statistical patterns learned during training, a RAG-based system works by first retrieving relevant content from a defined set of documents provided by the user, and then using that retrieved content as the direct basis for generating a response (Gupta, 2025). This means the AI is no longer guessing based on general knowledge. It is working directly from the actual materials that were given to it, which significantly

reduces the chance of producing inaccurate or fabricated content. Research supports this, with studies showing that RAG systems can meaningfully reduce hallucination rates compared to standard language model outputs, making them a far more dependable option for applications where factual accuracy is critical (Ovadia et al., 2024; Wang & Katsaggelos, 2025). For educational assessment specifically, RAG offers a practical way to ensure that AI-generated exam questions stay within the boundaries of what was actually taught. By restricting the AI to only the course materials an instructor uploads, such as lecture slides, handouts, or video transcripts, the system prevents outside or unverified information from entering the question generation process (Gupta, 2025). This boundary is exactly what general-purpose tools like ChatGPT and Gemini lack when used without structured restrictions. It is worth noting that RAG is not a perfect solution. Hallucinations can still occur if the system retrieves the wrong section of a document or misinterprets the retrieved content, which is why human review of generated items remains an important step (An et al., 2025; Gupta, 2025). Despite this, the grounded nature of RAG represents a significant improvement over unrestricted AI generation for high-stakes academic use (Mohammed et al., 2025). Nexam applies this approach by using RAG to ground every suggested exam item primarily in the instructor's uploaded course materials, restricting the generation process to the content the instructor has provided rather than drawing from general internet knowledge. Because RAG is not a perfect solution and hallucinations can still occur in edge cases, the system requires the instructor to review and approve every generated item before it is saved to the question bank or used in any exam. This human-in-the-loop design ensures that the efficiency gained from AI-assisted generation is balanced by the reliability of instructor oversight at every step.

Semantic Similarity Checking and Similarity Categories

Sentence-embedding methods represent the meaning of a sentence as a semantic vector that can be compared with other vectors using cosine similarity (Reimers & Gurevych, 2019). In Nexam, each newly drafted question and each approved question in the question bank are represented in this form. The comparison supports question-bank quality by detecting paraphrased or near-duplicate items that an exact keyword comparison may miss.

For system interpretation, the similarity result is grouped into two operational categories after the decision threshold has been validated during model evaluation. Similar refers to a pair whose cosine-similarity score is equal to or higher than the approved threshold; the item is flagged and sent to the instructor for verification before it can be saved. Far Similar refers to a pair whose score is below the threshold, indicating that the items are sufficiently different to continue through the ordinary instructor-review process. The categories are routing labels rather than automatic judgments of content quality, because the instructor still decides whether a question should be edited, approved, or rejected.

2.3 Synthesis

Several studies point to the need for better assessment practices and lighter workloads for teachers. DiDonato-Barnes, Fives, and Krause (2014) found that building a Table of Specifications (TOS) by hand often leads to errors and inconsistent test quality, while EDCOM 2 (2024) reports that teachers in the Philippines regularly work beyond 60 hours a week due to non-teaching tasks. These findings support the value of using AI-assisted tools to reduce repetitive assessment workflows, protecting both

teaching quality and teacher well-being (TNTP, 2024; Gallup & Walton Family Foundation, 2025).

Artificial intelligence has been shown to help address these problems. Teachers who use AI tools regularly save meaningful amounts of time on preparation and administrative work, freeing them to focus more on their students (Gallup & Walton Family Foundation, 2025; Steiner et al., 2023). Research also shows that AI can improve how test items are aligned to Bloom's Taxonomy, making assessments more consistent and objective than manual classification allows (Akdeniz et al., 2025; Banujan et al., 2023).

Retrieval-Augmented Generation (RAG) strengthens reliability by keeping AI outputs grounded in the instructor's course materials, reducing the risk of inaccurate or fabricated content (Gupta, 2025; Ovadia et al., 2024). Sentence-embedding methods complement this grounding by allowing question meaning to be compared through cosine similarity, supporting the detection of paraphrased or near-duplicate items before they enter the question bank (Reimers & Gurevych, 2019).

Taken together, these findings support the development of Nexam as an integrated platform that combines TOS-guided exam planning, course-material-grounded question drafting, semantic similarity checking, mandatory instructor review, centralized question-bank management, print-ready and LMS-compatible output, and mobile OMR checking. The literature therefore supports both the need for the system and the design choices used to connect assessment planning, content reliability, duplicate control, delivery, and scoring in one workflow.

2.4 Related Systems

The existing platforms that provide digital assessments, AI-driven question generation, and test bank management create the need for a solution that combines structured curriculum planning and fact-grounded AI generation into one accessible workflow. The existing systems examined below display both their positive attributes and the essential limitations that the proposed integrated platform must resolve.

Canvas and Moodle

Canvas and Moodle provide robust features that enable users to conduct quizzes, perform automatic grading, and manage class records efficiently. These platforms achieve their strongest functionality through digital assessment delivery and learning management capabilities. However, the testing process still requires manual configuration, as instructors must either create questions individually or import them from external test banks. While they effectively support assessment administration, they do not provide a native Table of Specifications (TOS) generation feature, Retrieval-Augmented Generation (RAG)-based question creation, or automated Bloom's Taxonomy alignment for learning outcome-based assessment design. Additionally, although Canvas and Moodle support online assessment and grading, they do not natively provide Optical Mark Recognition (OMR)-based answer sheet scanning and checking for printed examinations through an integrated mobile application.

Although Canvas and Moodle are effective for course delivery, online quizzes, automatic grading, and class-record management, their assessment-building workflow still depends heavily on manual configuration. Nexam addresses this limitation by connecting syllabus-driven TOS preparation, question drafting grounded in

instructor-provided materials, semantic similarity checking, instructor review and approval, print-ready and LMS-compatible outputs, and mobile OMR checking. This integrated workflow gives Nexam an advantage for institutions that need to manage both printed and digital examinations without moving between several separate tools.

The screenshot shows the Canvas LMS interface for creating a quiz. On the left is a navigation menu with options like Home, Announcements, Assignments, Discussions, Grades, People, Pages, Files, Syllabus, Outcomes, Quizzes, Modules, Conferences, Collaborations, Chat, New Analytics, Class Notebook, Accessibility Report, Zoom, YuJa Media, Echo360 ALP, and Settings. The main area is titled 'First Quiz' and includes a 'Quiz Instructions' text editor. Below this, there are settings for 'Quiz Type' (Graded Quiz), 'Assignment Group' (Assignments), and 'Options'. The options include checkboxes for 'Shuffle Answers', 'Time Limit' (with a minutes input), 'Allow Multiple Attempts', 'Let Students See Their Quiz Responses', 'Only Once After Each Attempt', 'Let Students See The Correct Answers', and 'Show one question at a time'. There are also input fields for 'Show Correct Answers at' and 'Hide Correct Answers at'.

Figure 3 Quiz Creation Interface of the Canvas Learning Management System
(University of Kentucky College of Arts & Sciences, 2024)

The screenshot shows the Moodle site interface. The top bar includes 'My new Moodle site', 'USERS', 'STORAGE', and a user profile for 'Olga Kokoulina'. The left sidebar contains a menu with Home, Dashboard, Calendar, Private files, My courses, Workplace Safety, Sample Course, Introduction to Moodle, and Site administration. The main content area is titled 'Course format' and shows settings for 'Format' (Topics format), 'Number of sections' (set to 1), 'Hidden sections', and 'Course layout'. A dropdown menu for 'Number of sections' is open, showing options from 0 to 19. The 'Hidden sections' setting is set to 'sections are shown in collapsed form', and the 'Course layout' setting is set to 'all sections on one page'.

Figure 4 Quiz Configuration Interface of the Moodle Learning Management System
(iSpring Solutions, 2024)

OpenAI ChatGPT (Used as an AQG Tool)

Educators frequently utilize general Large Language Models (LLMs) such as ChatGPT to rapidly create multiple-choice questions and other assessment items. The platform's primary strength lies in its high-speed text generation capabilities, enabling instructors to produce content efficiently. However, reliance on general-purpose LLMs introduces the risk of AI hallucinations, where generated content may contain inaccuracies or information not directly supported by instructional materials. While users can upload documents that partially support Retrieval-Augmented Generation (RAG), the system does not strictly constrain question generation to a teacher's syllabus, learning outcomes, or Table of Specifications (TOS). Furthermore, ChatGPT generates assessment content as raw text output, requiring instructors to manually organize, format, validate, and align questions for both printed examinations and Learning Management System (LMS) deployment. In addition, the platform does not provide built-in support for Optical Mark Recognition (OMR)-based answer sheet generation, scanning, or automated checking of printed examinations, requiring educators to rely on external tools and manual workflows for assessment evaluation.

ChatGPT can help instructors create questions quickly, but its output still requires the instructor to organize the items, align them with a TOS, check for repeated questions, format the examination, and use another tool for printed-answer-sheet checking. Nexam provides a more controlled workflow because questions are drafted

from instructor-provided course materials, checked for semantic similarity, reviewed and approved by the instructor, stored in a centralized question bank, and converted into print-ready or LMS-compatible outputs. Mobile OMR checking also allows the same system to process the results of printed examinations.



Figure 5 Generative AI Prompt Interface of OpenAI ChatGPT (UX Planet, 2023)

Google NotebookLM

Google NotebookLM functions as an advanced AI-powered research and knowledge synthesis assistant designed to ground its responses strictly on user-uploaded documents. Its primary strength lies in its Retrieval-Augmented Generation (RAG) architecture, which significantly reduces the risk of AI hallucinations by limiting generated content to information contained within the provided sources. This makes NotebookLM highly effective for summarization, content exploration, and document-based question answering. However, it is fundamentally a research and note-taking tool rather than an assessment generation platform. The system does not

provide a Table of Specifications (TOS) framework, automated item distribution based on instructional hours, Bloom's Taxonomy alignment, or examination blueprint generation. Furthermore, it lacks support for exporting examinations into Learning Management System (LMS) formats such as GIFT or QTI and does not offer Optical Mark Recognition (OMR)-based answer sheet generation, scanning, or automated checking capabilities. As a result, educators must rely on external tools to create, administer, and evaluate assessments derived from the generated content.

NotebookLM is valuable because it grounds responses in uploaded sources, but it remains primarily a research and synthesis tool. Nexam applies the same principle of source grounding within a complete assessment workflow by using the syllabus and course materials to guide TOS preparation and question drafting. Approved questions are stored in a reusable question bank and can be assembled into print-ready or LMS-compatible examinations, while mobile OMR checking supports the scoring of printed answer sheets.

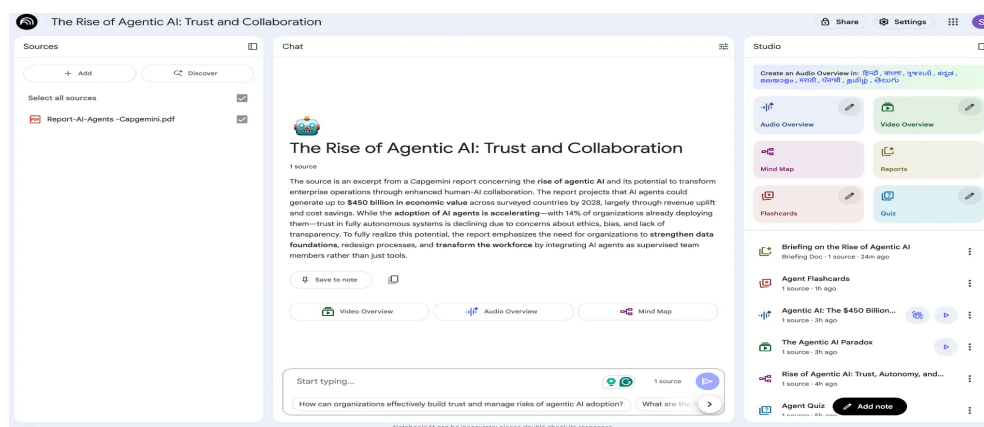


Figure 6 Document Upload and Synthesis Interface of Google NotebookLM (Analytics Vidhya, 2025)

MagicSchool AI

MagicSchool AI operates as a specialized educational generative tool designed to assist teachers in creating quizzes, instructional materials, and classroom resources through a simple prompt-based interface. The platform offers a user-friendly experience tailored to educational settings, enabling educators to generate assessment items

quickly while attempting to align content with general educational standards and learning objectives. However, the system does not implement a formal Table of Specifications (TOS) framework that mathematically distributes examination items according to instructional hours, topic weights, and Bloom's Taxonomy levels before question generation. Additionally, its export capabilities are primarily text-based or limited to selected integrations, lacking comprehensive dual-format export support for both print-ready examinations and Learning Management System (LMS) deployment. Furthermore, MagicSchool AI does not provide Optical Mark Recognition (OMR)-based answer sheet generation, scanning, or automated checking functionalities, requiring educators to use separate assessment and grading tools for printed examinations.

MagicSchool AI offers teachers a convenient way to generate educational content, but it does not begin with a formal TOS that distributes items according to topic weight and cognitive level. Nexam places this planning step at the beginning of the process, then connects it with grounded question drafting, similarity checking, instructor approval, dual-format examination output, and mobile OMR checking. As a result, the generated questions remain connected to the assessment blueprint and the instructor's final decisions.

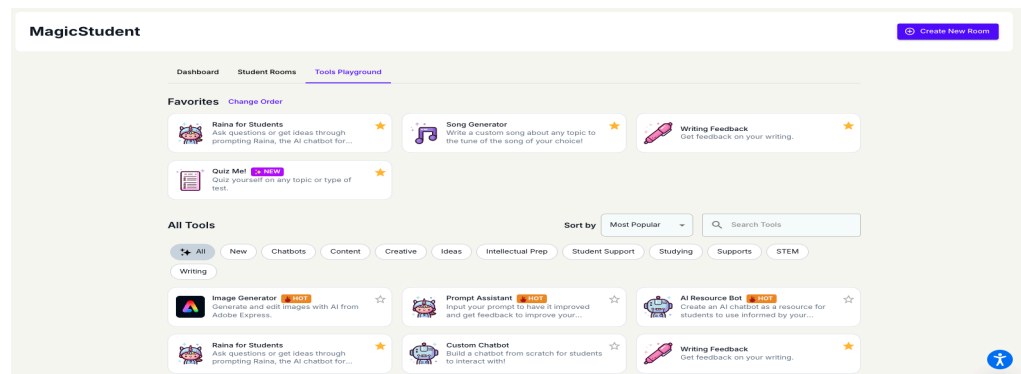


Figure 7 Educational Generation Interface of MagicSchool AI (MagicSchool AI, 2024)

QuestionWell

QuestionWell is a modern AI-powered assessment tool designed to transform reading materials into educational questions and activities. Its primary strength lies in its ability to rapidly generate assessment items and export them directly to various educational platforms such as Canvas, Quizizz, and Kahoot, making deployment highly convenient for educators. While its integration and export capabilities are notably strong, the platform functions primarily as a question-generation tool rather than a comprehensive assessment planning system. It does not provide a built-in Table of Specifications (TOS) framework that systematically distributes examination items according to instructional hours, topic weights, and Bloom's Taxonomy levels before question generation. Consequently, educators cannot ensure that assessments are mathematically balanced across cognitive domains during the test construction process. Furthermore, QuestionWell does not support Optical Mark Recognition (OMR)-based answer sheet generation, scanning, or automated checking for printed examinations, limiting its functionality to digital assessment creation and deployment rather than end-to-end examination management.

QuestionWell is particularly useful for generating questions quickly and exporting them to digital platforms. Nexam extends beyond this function by computing a syllabus-driven TOS before question generation, checking new items against approved questions for semantic similarity, requiring instructor review, and maintaining a centralized question bank. It also produces print-ready examination sets and supports mobile OMR checking, allowing the assessment process to continue from planning and question creation through printed-exam scoring.

The screenshot displays the QuestionWell web interface. At the top, there is a navigation bar with links for 'ate', 'Tools', 'History', and 'Pricing', along with an 'Upgrade' button and a user profile icon. Below the navigation bar, there are two buttons: 'Generate' and 'Import'. The main form allows users to specify the number of questions (10), the question type (Multiple Choice), the topic (The structure of an atom), the language (English), and the reading level (8th Grade). There is also a section for 'based on' with various icons and a word count (0/1,000 words). A 'Question Types' dropdown menu is visible. A large blue button labeled 'Generate Questions' is prominently displayed. At the bottom, there is a link to 'OR browse more AI tools for teachers'.

Figure 8 Educational Assessment Application Profile of QuestionWell (Clever, 2026)

ExamView Assessment Suite

ExamView serves as a traditional enterprise-grade assessment generation platform widely used by textbook publishers, educational institutions, and universities for creating and managing examinations. Its primary strength lies in its robust dual-format deployment capabilities, allowing educators to generate professionally formatted printed

examinations and export digital assessment files directly to supported Learning Management Systems (LMS). This flexibility makes ExamView highly effective for both paper-based and digital assessment environments. However, the platform relies entirely on static, pre-existing test banks and instructor-authored content. It does not incorporate generative artificial intelligence technologies that can automatically create new assessment items from instructional materials or dynamically generate questions based on learning outcomes. Furthermore, ExamView does not provide automated Table of Specifications (TOS) generation, Bloom's Taxonomy-based item distribution, or syllabus-driven assessment planning. Additionally, while it supports examination creation and deployment, it does not include Optical Mark Recognition (OMR)-based answer sheet generation, mobile scanning, or automated checking functionalities for printed examinations. As a result, educators must rely on separate OMR systems or manual checking processes when evaluating paper-based assessments.

ExamView is effective for formatting and distributing printed and digital tests, but it mainly depends on static or instructor-authored test banks. Nexam strengthens this workflow through assisted TOS preparation and RAG-based question drafting from instructor-uploaded materials. It also includes semantic duplicate detection, instructor approval, reusable question-bank management, dual-format output, and mobile OMR-based scoring, allowing assessment planning, production, and checking to remain within one connected system.

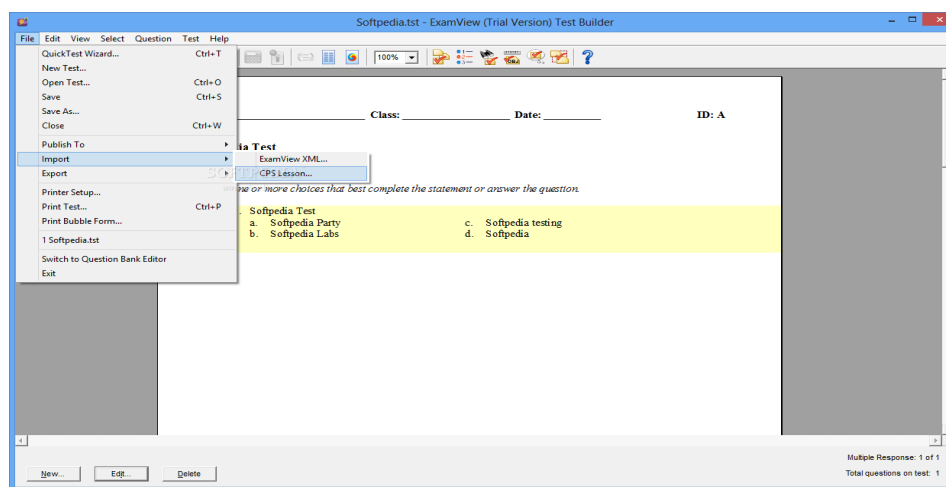


Figure 9 Test Document Builder of the ExamView Assessment Suite
Professional Edition (Softpedia, 2024)

Table 1 Comparison Table for Existing System and Proposed System

System/ Platform	TOS-Guided Item Distribution	RAG-Based Item Generation	Assisted Bloom's Alignment	Dual-Format Export (Print & LMS)	OMR-Based Answer Sheet Checking
Canvas/ Moodle	No	No	No	Partial	No
OpenAI ChatGPT	Partial	Partial	Partial	No	No
Google NotebookLM	No	Yes	No	No	No
MagicSchool AI	No	Partial	Partial	Partial	No
QuestionWell	No	Partial	Partial	Yes	No
ExamView Suite	No	No	No	Yes	No
Nexam (Proposed System)	Yes	Yes	Yes	Yes	Yes

A review of existing educational assessment platforms reveals a substantial deficiency in the current market: no single system successfully integrates the entire assessment creation pipeline from initial curriculum planning to final deployment.

Current technologies fragment the workflow into isolated steps. Learning Management Systems (Canvas, Moodle) and traditional assessment suites (ExamView) enable excellent exam distribution and dual-format deployment, but they require excessive manual labor from educators to configure a Table of Specifications (TOS) and write the actual questions. Conversely, the rapid item-generation features of modern AI tools (ChatGPT, MagicSchool AI, QuestionWell) fail to provide a complete solution because they either lack strict factual grounding, struggle with precise pedagogical alignment, or completely lack formal, mathematical TOS planning prior to generation. While advanced tools like NotebookLM solve the issue of AI hallucination through strict Retrieval-Augmented Generation (RAG), they are entirely disconnected from standard assessment deployment formats.

Nexam addresses this fragmentation by integrating an assisted TOS builder, course-material-grounded question drafting, semantic similarity checking, mandatory instructor approval, a centralized question bank, print-ready and LMS-compatible outputs, and mobile OMR checking in one workflow. Its main advantage is end-to-end continuity: the syllabus, TOS, approved questions, examination files, answer keys, and scoring results remain connected. Instructors retain final authority over every item, while the system reduces repetitive work, flags duplicates, preserves traceability to source materials and the TOS, and supports both paper-based and LMS-based assessment without relying on separate tools.

CHAPTER III

MATERIALS AND METHODS

3.1 Software Methodology

This study adopts the Iterative Development Model guided by the PADIM framework, covering the phases of Planning, Analysis, Design, Implementation, and Maintenance, as the software development methodology for designing, developing, and evaluating Nexam. In this combined approach, the PADIM framework structures each individual iteration cycle, ensuring that every increment passes through a consistent sequence of planning, analysis, design, implementation, and maintenance before the next cycle begins. This approach is most appropriate for Nexam because the system is composed of interdependent modules that cannot be safely developed or validated simultaneously without significant risk. Alshamrani and Bahattab (2015) classify the iterative model as ideal for complex projects where it is highly risky to develop the whole system at once, as risk is distributed across smaller, controlled increments rather than concentrated in a single large-scale deployment.

The iterative model is further justified by Nexam's reliance on non-deterministic AI components, particularly its Retrieval-Augmented Generation (RAG) pipeline and large language model (LLM) question generation engine, which cannot be fully validated through a single development pass. Westacott et al. (2023) confirm that Automated Item Generation (AIG) software requires an iterative development process to continuously refine the parameters governing AI output, while the Large Language Models Hallucination Survey (n.d.) establishes that hallucination mitigation strategies, such as supervised fine-tuning and self-refinement, are inherently iterative processes. By organizing development into structured iterations, with each iteration following the

PADIM phases to ensure systematic progression from requirements gathering through to maintenance and review, Nexam's development progressively validates every module before full integration. This ensures that the final platform is stable, curriculum-aligned, and ready for ISO/IEC 25010 user evaluation exclusively with university instructors.

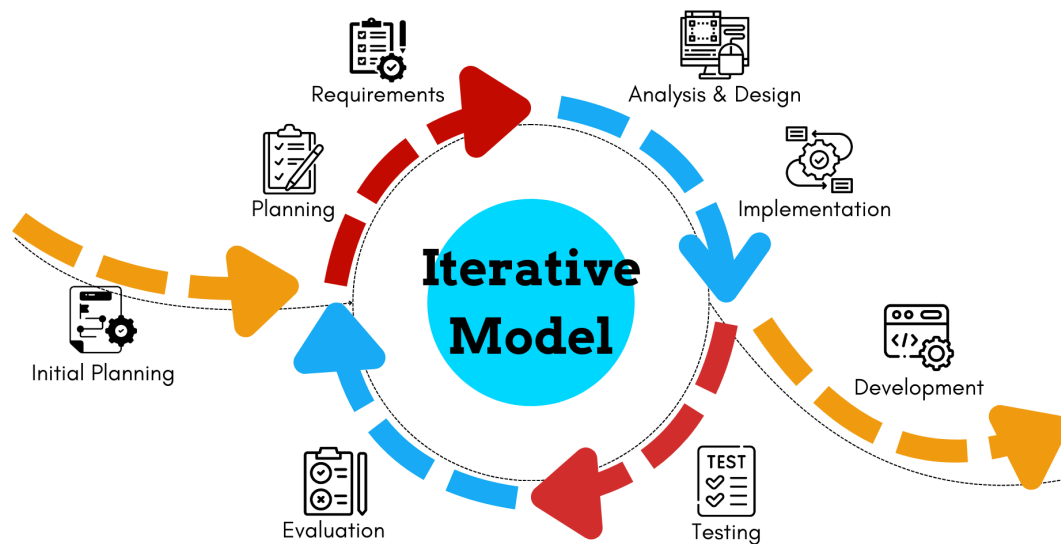


Figure 10 The Iterative Model (Nguyen, 2024) .

Algorithm Development and Process Flow

To ensure rigorous validation of the system's core logic and to mitigate AI hallucinations, the development is divided into explicit algorithmic pipelines.

1. Retrieval-Augmented Generation (RAG) and Semantic Retrieval Algorithm To ensure the reliability and factual accuracy of the AI, question generation is restricted entirely to instructor-uploaded materials.

- **Data Ingestion:** Extracted instructional text from the syllabus and supplementary files is split into overlapping semantic chunks.
- **Vectorization:** These text chunks are converted into high-dimensional vector embeddings and securely stored in a dedicated vector database, which operates alongside the primary MySQL relational database used for storing structured data like instructor credentials and exam rules.
- **Semantic Retrieval:** The system queries the vector space using the specific TOS topic to retrieve only the most relevant text chunks. These chunks constrain the Google Gemini API, ensuring drafted questions are grounded in verified academic context.

2. Semantic Similarity (Deduplication) Algorithm To maintain the integrity of the question bank, newly drafted questions are vectorized, and their cosine similarity is calculated against existing approved items. Items exceeding the predefined similarity threshold are automatically flagged as duplicates to prevent exam redundancy.

3. Mobile Optical Mark Recognition (OMR) Algorithm The mobile grading component utilizes computer vision algorithms to process printed answer sheets. The algorithm detects corner anchors on the printed OMR sheet to calculate perspective transformation and deskew the captured image. It then isolates response contours and calculates pixel density to determine the student's marked answers, cross-referencing them against the generated answer key to compute the final score.

The Process of the Table of Specifications (TOS)

The system features an Automated TOS Weighting Algorithm that replaces manual, error-prone computations with a deterministic mathematical process. The step-by-step process flow for generating the TOS blueprint is as follows:

1. **Parameter Extraction:** The system parses the uploaded syllabus to extract the fundamental parameters: the list of course topics and the designated instructional hours for each topic.
2. **Weight Computation:** The algorithm calculates the proportional weight of each topic. It divides the topic's instructional hours by the total instructional hours of the course, multiplying by 100 to yield the percentage weight.
3. **Item Allocation:** Based on the instructor-defined total number of exam items, the system allocates the exact number of questions per topic by multiplying the total exam items by the computed topic weight.
4. **Bloom's Taxonomy Distribution:** Finally, the algorithm maps the allocated topic items across the six cognitive levels of the Revised Bloom's Taxonomy. The system distributes the items based on the cognitive percentage weights specified by the instructor, locking these parameters as strict constraints for the AI question generator.

AI Accuracy, Reliability, and Evaluation Matrix

To scientifically validate the reliability and accuracy of the system's AI components specifically its ability to correctly classify generated questions into the appropriate cognitive levels of Bloom's Taxonomy the algorithm will be evaluated using standard machine learning classification metrics.

A dataset of AI-generated questions will be manually validated by subject matter experts to establish a "ground truth." The system's algorithmic predictions will then be compared against this ground truth using a Confusion Matrix. The evaluation will utilize the following standard metrics:

- **Accuracy:** Measures the overall proportion of correctly classified cognitive levels across all generated questions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

equation 1

- **Precision:** Measures the reliability of a specific classification, calculating how many of the questions assigned to a specific cognitive level actually belong to that level.

$$Precision = \frac{TP}{TP + FP}$$

equation 2

- **Recall:** Measures the algorithm's ability to successfully identify all actual questions belonging to a specific cognitive level.

$$Recall = \frac{TP}{TP + FN}$$

equation 3

- **F1-Score:** The harmonic mean of Precision and Recall, providing a balanced and comprehensive metric for uneven class distributions.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

equation 4

(Where TP = True Positive, TN = True Negative, FP = False Positive, and FN = False Negative).

3.2 System Planning

The planning stage provides the foundation for the entire project. The development team will plan Nexam, A RAG and OMR-Based System for Syllabus-Driven TOS Configuration and Answer Sheet Checking after identifying the problems at higher education schools in Mati City through initial discussions with university instructors. The preparation includes identifying the primary user role (strictly faculty), defining the scope and limitations specifically focusing on objective exam formats and offering both print-ready test outputs and LMS-compatible export files and selecting the Iterative Development Model guided by the PADIM framework to structure the work. The researchers will also identify the necessary software tools and technologies and secure the required permissions from university administration before advancing to the analysis phase.

Development is broken down into smaller, manageable iterations, each focusing on specific modules of the system, such as the assisted TOS configuration module, the AI-assisted question generation module, and the mobile OMR scanning module. The planning phase also includes establishing the development environment, identifying the programming frameworks and database architecture needed for both the web and mobile interfaces, and creating a detailed project schedule to ensure completion within the target timeline.

Feasibility Study

The feasibility study evaluates whether the proposed system can be built, implemented, and maintained using the available resources and time limits. This assessment covers three parts: operational feasibility, which examines if the system will work well in the university setting; economic feasibility, which analyzes the costs and benefits of the project; and technical feasibility, which checks if the needed technology and AI skills are present. These parts help determine if the system is possible, sustainable, and beneficial for local schools.

Operational Feasibility

The operational feasibility evaluates whether the system can be effectively utilized by its target users, specifically higher education instructors. The system is designed to reduce the administrative effort required in preparing exams and to support a more consistent exam structure by providing a centralized platform for creating syllabus-based, Table of Specifications (TOS)-aligned assessments. As a web-based architecture, instructors can manage the entire exam creation process through a standard web browser, ensuring high accessibility.

The platform directly addresses common challenges in exam preparation, such as manually calculating item distributions per topic and ensuring that questions are factually grounded in actual course content. It drafts objective questions from instructor-provided materials and organizes them according to the TOS blueprint, while strictly requiring instructors to review, edit, and approve all questions before they enter the centralized bank. The system also supports dual-format delivery, providing flexibility to export exams as print-ready offline papers (paired with OMR sheets) or as digital files

formatted for Learning Management Systems. Ultimately, the system integrates seamlessly into current pedagogical workflows and provides highly practical, time-saving support for faculty, making it suitable for broad institutional adoption.

Economic Feasibility

Table 2 Estimated Implementation cost of Nexam

Cost Category	Phase	Estimated Cost (₱)
A — Development Costs (In-house / Capstone Team)		
1. Planning	P	-
2. Analysis	A	-
3. Design	D	-
4. Implementation (Iterations 1–6)	D	-
B — Software & Infrastructure Costs		
Cloud hosting (e.g., Railway / Vercel / Render)	I	₱6,000.00
Database service (MySQL Cloud Hosting)	I	₱7,344.00
Vector database service (for RAG pipeline)	I	₱4,800.00
Domain name & SSL certificate — annual	I	₱2,000.00
Google Gemini API usage — annual estimate	I	₱8,400.00
C — Deployment & Training Costs		
Deployment & data migration	I	₱2,000.00
User training — instructors & students	M	₱3,000.00
User evaluation & acceptance testing (ISO/IEC 25010)	M	₱1,500.00
D — Maintenance Costs (Annual, Post-launch)		
Development & distribution of user manual	M	₱2,000.00
Monthly iteration reviews & minor system updates	M	₱6,000.00

AI model prompt retraining & RAG pipeline updates	M	₱3,000.00
Data Privacy Act (RA 10173) compliance review & documentation	M	₱0.00 (handled by university Data Protection Officer)
Total Estimated Cost		₱46,044.00

The total estimated cost of the Nexam system amounts to ₱46,044.00, which falls within the target operational budget range of ₱30,000.00 to ₱50,000.00. The cost breakdown is organized across four major categories: development costs, software and infrastructure costs, deployment and training costs, and annual maintenance costs. Development labor costs are recorded as zero since the system is built entirely by the in-house capstone team, eliminating the need for contracted software developers. The most significant cost driver is the Google Gemini API usage fee, which supports Nexam's core RAG-based question generation pipeline throughout the deployment period. Data Privacy Act compliance costs are likewise recorded as zero, as this responsibility falls under the jurisdiction of the university's designated Data Protection Officer in accordance with RA 10173.

Table 3 Cost and Benefits of Nexam

Benefit Category	Basis / Computation	Amount (₱)
COST		
Total Cost		₱46,044.00
BENEFITS		
Total Hours Saved Annually	1,230 hrs × ₱1,395.68/hr	
Monthly Savings	₱176,154.55 ÷ 12 months	₱14,679.55
Annual Savings	₱1,716,686.40 ÷ 12 months × 12	₱176,154.55

Reduced photocopying errors & reprint costs	Semantic deduplication prevents repeated items, eliminating the need to manually review and rebuild question banks each semester	₱10,000.00
Reduced manual TOS computation errors & rework	Assisted TOS Builder	₱10,000.00
Improved compliance & audit readiness	Auto-generated TOS alignment reports per CHED CMO No. 4 s. 2020	₱8,000.00
Faster and more accurate exam construction	AI-assisted RAG generation reduces the manual effort of item writing	₱18,000.00
Total Benefits		₱176,154.55
Return on Investment & Payback Period		
Metric	Computation	Result
Return on Investment (ROI)	$\frac{((\text{Net Benefits} - \text{Total Cost}) \div \text{Total Cost})}{(\text{₱176,154.55} - \text{₱46,044.00}) \div \text{₱46,044.00} \times 100}$	282.61%
Payback Period	$(\text{Total Cost} \div \text{Monthly Savings})$ $\text{₱46,044.00} \div \text{₱14,679.55}$	3.14 months

In crafting the Cost-Benefit Analysis, man-hours were carefully estimated to quantify the potential savings generated by the system. The analysis assumed that Nexam could meaningfully reduce instructors' manual workload in exam construction, specifically in TOS item distribution calculation, question drafting and formatting, print-ready exam set preparation, question bank deduplication and review, and TOS alignment report generation. These are the five processes directly addressed by the system's modules, as reflected in the Time Savings Analysis table above. System-assisted grading applies only to digital exam submissions, and since the system is designed for dual-format delivery where printed exams are also produced, manual grading of printed papers using the system-generated answer keys remains the instructor's responsibility and is not included in the savings estimate. Based on estimated time savings of seven (7) hours per exam cycle across 15 instructors, using an average daily rate of ₱30,705.00 divided by 22 working days, the equivalent hourly

rate was computed and multiplied by the saved hours, resulting in monthly savings of ₱14,679.55 and an annual benefit of ₱176,154.55. Combined with additional quantified benefits from reduced rework costs, improved CHED compliance readiness, and faster AI-assisted exam construction, the total annual benefits confirm that Nexam delivers measurable institutional value well beyond its total estimated cost of ₱46,044.00, yielding a return on investment of 282.61% and a payback period of approximately 3.14 months.

Table 4 Time Saving Analysis of Nexam

Time Savings Analysis

Process	Current Manual Time	Assisted Time	Time Saved per Transaction	Annual Transactions	Total Hours Saved
TOS construction & item distribution calculation	3 hrs/exam	10 mins/exam	2.83 hrs	120 exams	339.60 hrs
Exam question writing & formatting	4 hrs/exam	30 mins/exam	3.50 hrs	120 exams	420.00 hrs
Print-ready exam set preparation (Sets A & B)	2 hrs/exam	5 mins/exam	1.92 hrs	120 exams	230.40 hrs
Question bank deduplication & review	1 hr/exam	Automatic	1.00 hr	120 exams	120.00 hrs
TOS alignment report generation	1 hr/exam	Automatic	1.00 hr	120 exams	120.00 hrs
Total Annual Savings					1,230.00 hrs

The economic feasibility analysis confirms that Nexam is a financially sound and sustainable investment for institutional adoption. With a total estimated cost of only ₱46,044.00, driven primarily by the Google Gemini API and cloud infrastructure fees while keeping development labor costs at zero through the in-house capstone team, the

system delivers annual benefits totaling ₱176,154.55 through instructor time savings, reduced rework costs, improved CHED compliance readiness, and faster AI-assisted exam construction. The resulting ROI of **282.61%** and a payback period of approximately **3.14 months** strongly affirm the system's financial viability. These figures demonstrate that Nexam is not only technically feasible but also economically justified and aligned with the fiscal priorities of higher education institutions in Davao Oriental.

Technical Feasibility

The technical feasibility of Nexam demonstrates that the system can be developed using currently available technologies and resources accessible to the development team. Development is executed on standard workstations with sufficient memory and processing power to build, compile, and test a full-stack web application.

The system utilizes a modern JavaScript-based technology stack. React provides the front-end interface, which is responsive and accessible across varying screen dimensions. Node.js handles all server-side processing, including file handling, text extraction, TOS calculations, similarity checking, exam assembly, and PDF generation. For data ingestion, pdf-parse extracts text from uploaded PDFs, and mammoth processes DOCX files. For website URLs, Axios fetches the page content and Cheerio parses the HTML. For YouTube links, a Node.js transcript library retrieves available captions and converts the content into clean text.

Because the system relies on a Retrieval-Augmented Generation (RAG) architecture, data storage utilizes a hybrid approach. MySQL serves as the primary relational database management system to securely store faculty credentials, TOS blueprints, structured question banks, and OMR exam results. Simultaneously, extracted

instructional text is chunked and converted into vector embeddings, which are stored in a dedicated vector database to power the AI semantic search. LangChain manages the interaction between this vector retrieval process and the Google Gemini API, which drafts the questions based on the retrieved content and the finalized TOS. PDF rendering for print-ready exam sets is managed by Puppeteer.

Finally, the mobile component utilizes native camera APIs and computer vision libraries to execute the Optical Mark Recognition (OMR) grading process on standard mobile devices. Overall, the selected frameworks, paired with the team's technical proficiency, confirm that the system can be successfully engineered and deployed within the scope of the study.

Work Breakdown Structure (WBS)

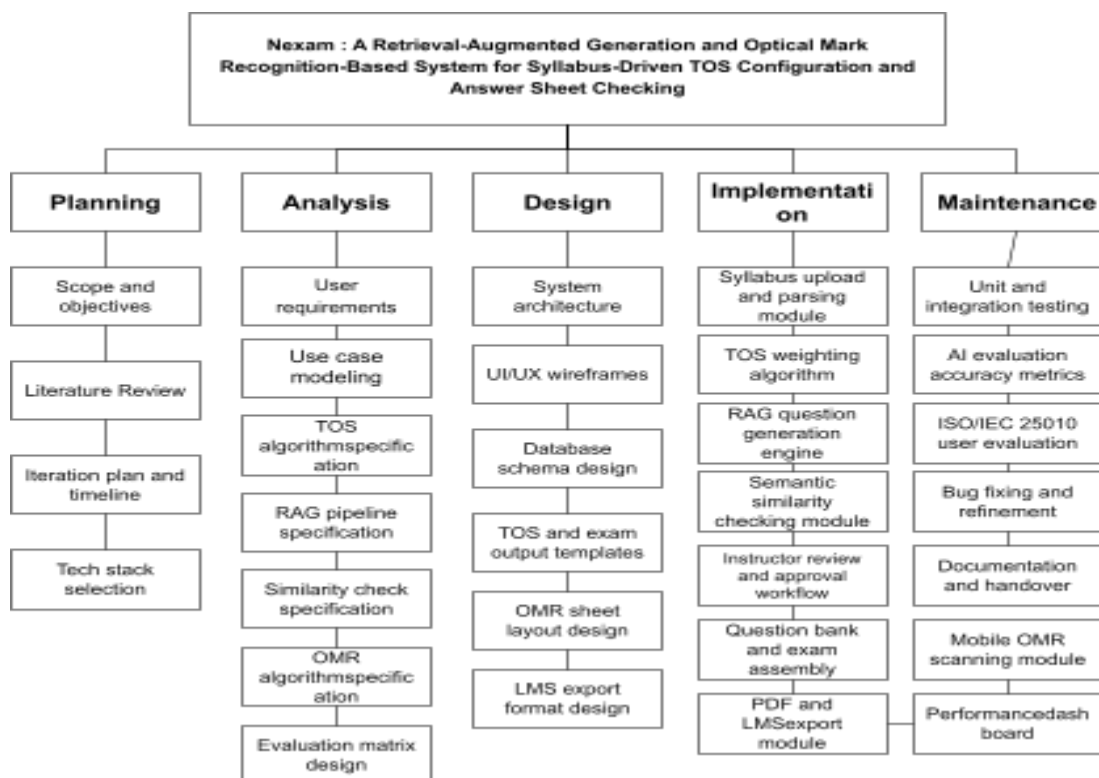


Figure 11 Work Breakdown Structure
Gantt Chart

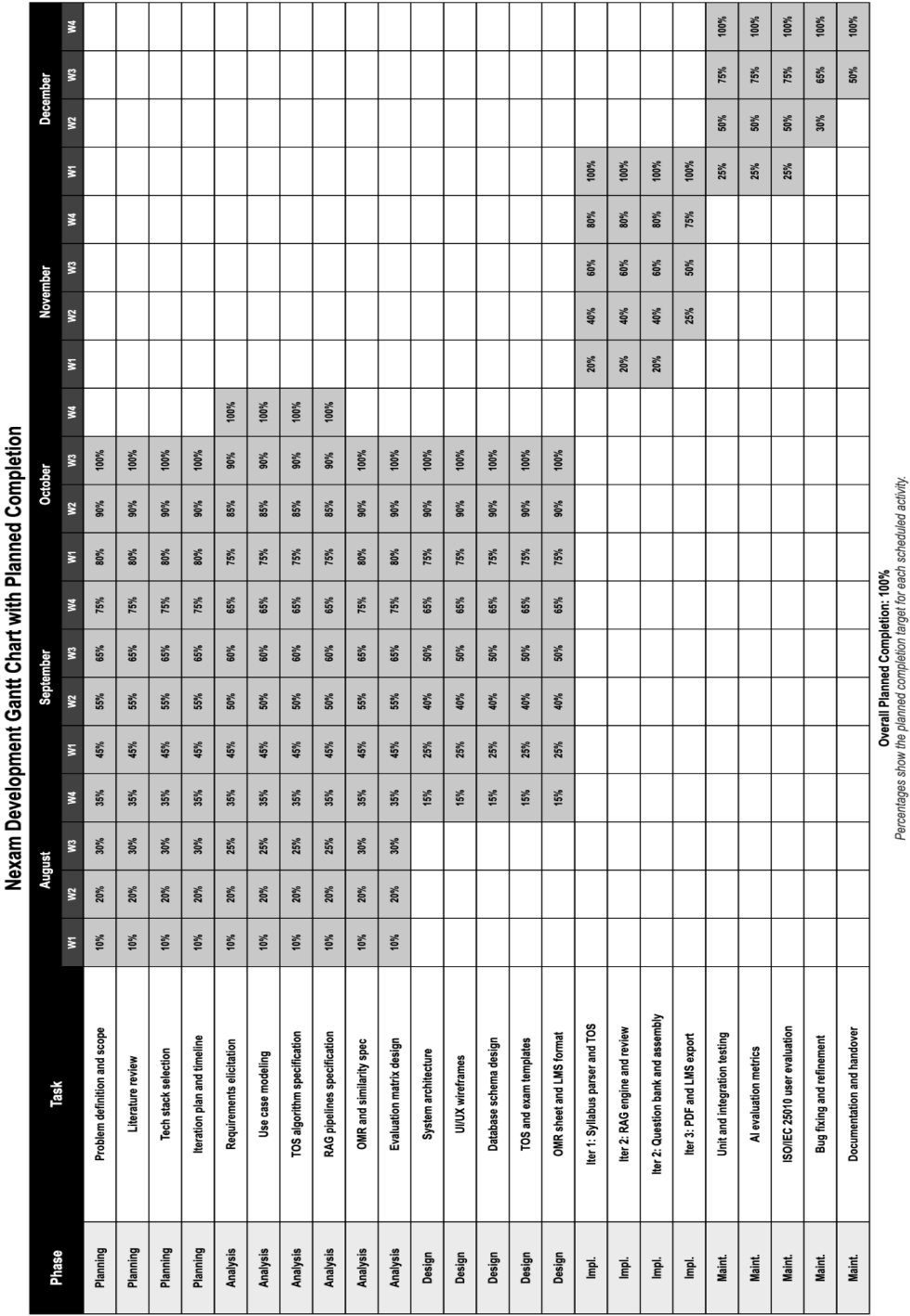


Figure 12 Gantt Chart

3.3 System Analysis

Higher education instructors in Mati City currently manage the entire exam construction process through a series of manual, disconnected steps. As illustrated in the Business Process Model and Notation of the current system, the process begins with the instructor reviewing the course syllabus and manually computing the Table of Specifications, which requires carefully calculating how many questions each topic should receive based on instructional hours and then distributing those items across the different levels of Bloom's Taxonomy. This computation is done entirely by hand, leaving significant room for miscalculation and inconsistency across different instructors and subject areas.

Once the TOS is drafted, the instructor manually writes the exam questions and checks whether they are properly aligned to the blueprint. If the questions do not satisfy the TOS requirements, the instructor must go back and rewrite or replace items, creating a repetitive loop that consumes considerable time before a satisfactory exam can be produced. Once alignment is confirmed, the instructor formats and prints the exam papers, distributes them to students, collects the submitted papers after the examination period, grades them manually, and then records and reports the results. Throughout this entire process, there is no centralized system connecting the TOS, the question bank, the printed exam, and the scoring. Each step is handled independently, meaning errors made in one stage are not automatically caught in the next. Students, on the other hand, simply receive whatever paper exam is distributed, answer the questions, submit their papers, and wait for results that depend entirely on how quickly the instructor can

complete manual grading. This fragmented and labor-intensive workflow is the core problem that Nexam is designed to help address. The system takes over the most repetitive and error-prone parts of the process — calculating item distributions, drafting questions from uploaded materials, checking for repeated items, and formatting exam outputs — while keeping the instructor in control of content review and final approval at every stage. The goal is not to remove the instructor from the process but to reduce the mechanical work so that instructors can spend more of their time on the parts of exam construction that genuinely require their professional judgment.

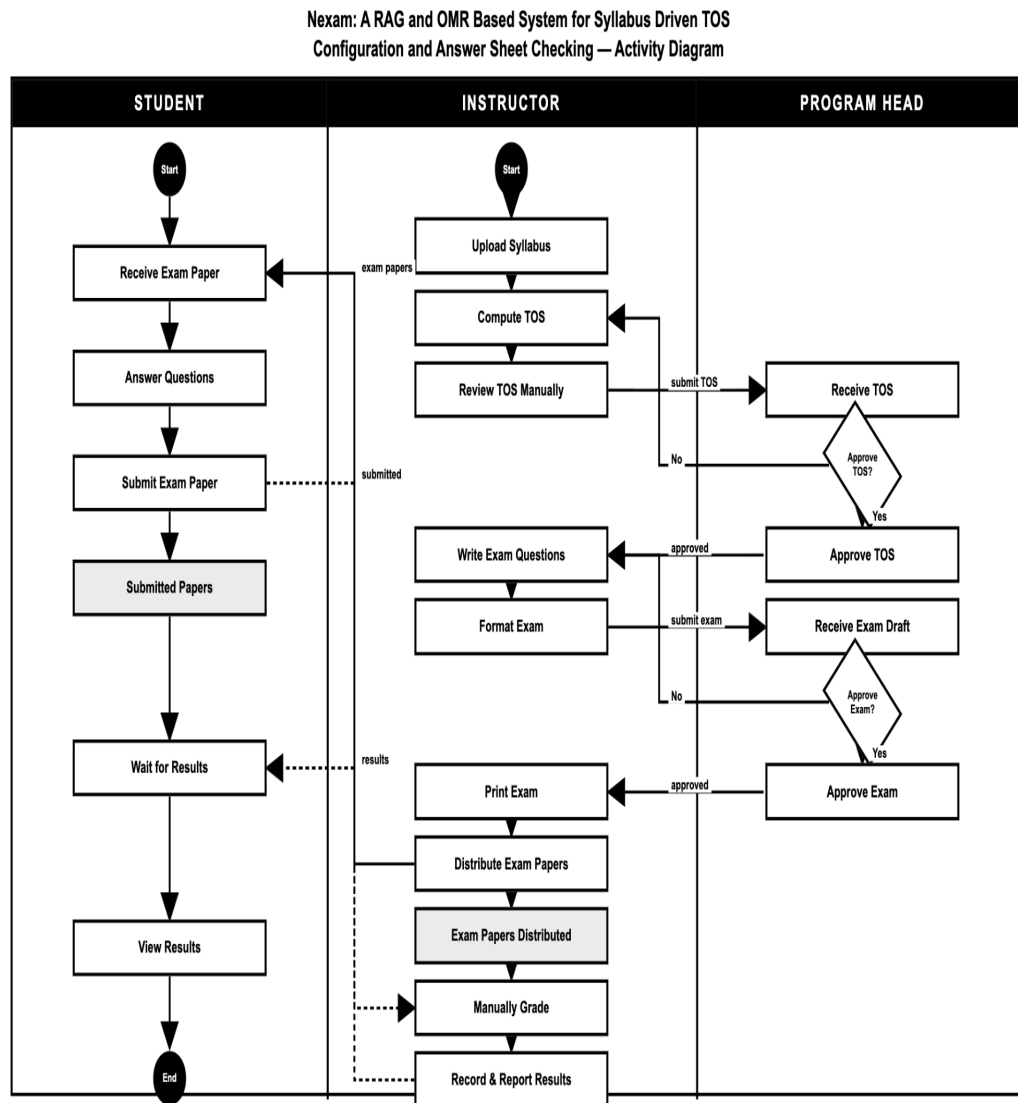


Figure 13 Business Process Model and Notation (BPMN) of the current system

Requirement Specification

The Requirement Specification defines the operational boundaries of Nexam, translating the study's general objectives into concrete technical mandates. These requirements ensure the platform effectively bridges the gap between manual TOS construction and AI-driven assessment.

1. Functional Requirements

The system shall provide the following functional capabilities:

- The system shall allow users to access the platform based on Role-Based Access Control (RBAC), strictly supporting a Faculty/Instructor role.
- The system shall allow instructors to:
 - Upload course syllabi (PDF/DOCX format) to extract topics and instructional hours for the automatic generation of the Table of Specifications (TOS).
 - Upload and manage supplementary course materials, including documents, presentation files, website links, and video transcripts, which serve as the primary source for AI-driven question generation.
 - Set up and refine the TOS by adjusting the system-calculated topic weights, item distributions, and Bloom's Taxonomy cognitive levels.
 - Review, edit, and manually approve AI-generated questions before they are stored in the question bank.
 - Show whether there are enough approved questions per topic and Bloom's level during exam setup.
 - Create print-ready test papers (e.g., Set A and Set B) with corresponding answer keys and TOS alignment reports.
 - Generate and print machine-readable OMR answer sheets uniquely paired to the generated physical exam sets.
 - Use the connected mobile application to scan completed OMR answer sheets for automated grading.

- View exam results through a web dashboard, including individual student scores, class performance summaries, and item-level results derived from the mobile OMR scans.
- The system shall:
 - Use Retrieval-Augmented Generation (RAG) to draft objective questions based strictly on the instructor's uploaded course materials.
 - Execute a Semantic Similarity Checking algorithm to detect and flag duplicate or near-duplicate questions before they are saved to the question bank.
 - Use the topic name entered in the TOS as a semantic search query to retrieve the most relevant content chunks from the instructor's uploaded materials before generating questions for that topic.
 - Tag every approved question with both its topic and its Bloom's Taxonomy cognitive level, and query the question bank using both tags when assembling an exam based on the finalized TOS blueprint.

2. Non-Functional Requirements

The system shall meet the following quality and performance standards:

- **Performance**
 - The system shall process and respond to user requests within a reasonable timeframe under normal load conditions.
 - The mobile application shall rapidly process Optical Mark Recognition (OMR) scans and synchronize the grading data to the web database.
- **Security**

- The system shall implement Role-Based Access Control (RBAC) to ensure that only authorized faculty can access the platform.
- The system shall ensure that all stored data, including exam questions, TOS configurations, and student OMR results, are encrypted and accessible only to the authorized instructor.
- **Reliability**
 - The system shall generate exam questions based *only* on the instructor's uploaded course materials to mitigate the risk of AI hallucinations or unrelated content.
 - The system shall mathematically verify that enough approved questions are available for each topic and Bloom's level before allowing an exam to be finalized.
- **Usability**
 - The system shall provide an intuitive web interface for instructors that does not require advanced technical training.
 - The mobile OMR scanner interface shall provide clear visual feedback to the instructor during the physical sheet scanning process.
- **Portability**
 - The web platform shall be accessible through standard web browsers on desktop and laptop devices.
 - The system shall support exporting item bank files in standard formats, specifically GIFT (for Moodle) and XML (for Canvas), for manual upload to supported learning management systems.
- **Scalability**

- The system shall handle increasing amounts of course materials, generated questions, and scanned OMR records while maintaining stable performance.
- **Maintainability**
 - The system shall be designed in modular components so that features (like the RAG pipeline or OMR scanner) can be updated independently.
- **Data Privacy and Compliance**
 - The system shall comply with the Philippine Data Privacy Act of 2012 (Republic Act 10173), ensuring that student performance records from OMR scans are handled securely.
 - The system shall support dual-format assessment delivery (print-ready papers and LMS exports) in line with flexible learning requirements under CHED CMO No. 4, s. 2020.

Proposed Solution

The proposed solution is the development of Nexam, an A RAG and OMR-Based System for Syllabus-Driven TOS Configuration and Answer Sheet Checking. This solution comprises a mobile-responsive web platform paired with a mobile Optical Mark Recognition (OMR) application, specifically designed to reduce the manual administrative effort involved in creating Tables of Specifications (TOS), drafting exam questions, and grading physical test papers.

The workflow begins with data ingestion. The system allows instructors to upload a course syllabus, from which it extracts the target topics and instructional hours to automatically compute the TOS blueprint, including percentage weights and Bloom's Taxonomy cognitive distributions. Instructors then upload supplementary course

materials such as documents, presentation files, website links, and video transcripts which serve as the strict, factual basis for AI-assisted question generation using a Retrieval-Augmented Generation (RAG) architecture.

Based on the TOS blueprint, the system generates a draft pool of objective questions. To ensure question bank integrity, a semantic similarity algorithm checks these drafts against existing items to flag duplicates. The unique questions are then presented to the instructor, who must manually review, edit, and approve each item before it is committed to the centralized question bank. This ensures that the final exam content remains entirely under instructor control.

For exam delivery, the system eschews online student test-taking in favor of a robust dual-format deployment. The platform generates print-ready physical test papers (e.g., Set A and Set B) complete with TOS alignment reports, answer keys, and machine-readable OMR answer sheets. Concurrently, the system allows instructors to export the approved question banks in GIFT and XML formats for seamless, manual integration into institutional Learning Management Systems like Moodle and Canvas.

Finally, the system addresses the grading burden. Following a physical examination, instructors utilize the Nexam mobile application to scan the students' completed OMR answer sheets via the device camera. The mobile OMR module computes the scores against the generated answer key and automatically synchronizes the results to the web platform, providing the instructor with instant access to individual scores, class performance summaries, and item-level analytics through a centralized performance dashboard.

Ethical Considerations

The development, evaluation, and deployment of Nexam are governed by strict principles of information technology ethics, algorithmic accountability, and data governance.

- **Selection of Participants**

Participants will be drawn from currently employed teaching faculty who handle at least one course requiring formal examinations during the evaluation period. Faculty members who are on leave, do not administer formal exams, or are unable to provide informed consent will be excluded from participation.

- **Benefits to Participants**

Participating faculty will gain early access to Nexam and directly experience reductions in the time and effort required for exam construction. Their feedback will also directly shape system improvements beyond the study period.

- **Risk Assessment**

Risks associated with participation are minimal. No sensitive personal data beyond institutional credentials and course materials will be collected. The primary identified risk—unintended disclosure of uploaded course materials—is mitigated through encryption and restricted access controls. Participants may withdraw at any point without consequence.

- **Data Storage and Confidentiality**

The system complies with the Philippine Data Privacy Act of 2012 (R.A. 10173). All data—including faculty credentials, uploaded course materials, and OMR-derived student records—shall be encrypted at rest and in transit and accessible only to authorized personnel. Evaluation data will be anonymized prior

to analysis, ensuring no individual participant is identifiable in any reported findings. Data lifecycle management will be coordinated with the university's designated Data Protection Officer.

- **Conflict of Interest**

The researchers declare no financial, commercial, or personal conflict of interest. Nexam was developed solely for academic research purposes with no external commercial funding. Any conflict of interest that arises will be disclosed immediately to the research adviser and the institutional review body.

- **Feedback and Dissemination**

Participants will receive a summary of evaluation findings upon conclusion of the study. Full research outputs will be submitted for publication through institutional repositories, journals, or conference proceedings, with all findings presented in aggregate form to protect participant confidentiality.